

Finding the Whole or the Percent Rate

Answer Key

Grade 7

Quick Answers

- | | |
|---------|--|
| 1. 35 | 8. 360 |
| 2. 75 | 9. 260 |
| 3. 48 | 10. 28.3 |
| 4. 60 | 11. (a) full price in dollars = 420, (b) amount still owed, in dollars = 336 |
| 5. 90 | 12. (a) number of students asked = 150, — |
| 6. 25 | |
| 7. 12.5 | |
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What is verified

11 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 7

6.RP.A.3.c – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 14 tagged checks.

1. 40 seats, 14 taken. What percent are taken?

Solution: The part is 14 and the whole is 40, so the rate is $\frac{14}{40} = 0.35$, and $0.35 \times 100 = 35$ percent. 35

2. 24 questions, 18 correct. What percent is correct?

Solution: $\frac{18}{24} = 0.75$, so the rate is $0.75 \times 100 = 75$ percent. 75

3. 12 tickets is 25% of the printing. How many were printed?

Solution: Here the part is known and the whole is missing. Twenty-five percent is 0.25, so $0.25x = 12$ and $x = 12 \div 0.25 = 48$ tickets. 48

4. 45 seedlings, 27 flowered. What percent flowered?

Solution: $\frac{27}{45} = 0.6$, and $0.6 \times 100 = 60$ percent.

60

5. 63 kilometres is 70% of the trail. How long is the trail?

Solution: The whole is missing again: $0.70x = 63$, so $x = 63 \div 0.70 = 90$ kilometres.

90

6. Solve $\frac{60}{100}x = 15$ for the class size.

Solution: $\frac{60}{100} = 0.6$, so the equation is $0.6x = 15$. Divide both sides by 0.6: $x = 15 \div 0.6 = 25$ students. Check: 60% of 25 is 15.

25

7. 56 marbles, 7 green. What percent are green?

Solution: $\frac{7}{56} = 0.125$, and $0.125 \times 100 = 12.5$ percent. A percent may be a decimal.

12.5

8. 45 books is 12.5% of the cart. How many books are on the cart?

Solution: 12.5% is 0.125, so $0.125x = 45$ and $x = 45 \div 0.125 = 360$ books.

360

9. Solve $\frac{35}{100}x = 91$ for the stock size.

Solution: $\frac{35}{100} = 0.35$, so $0.35x = 91$ and $x = 91 \div 0.35 = 260$ shirts. Check: 35% of 260 is 91.

260

10. 60 birds counted, 17 finches. What percent, to the nearest tenth?

Solution: $\frac{17}{60} = 0.28333\dots$, so the rate is $0.28333\dots \times 100 = 28.333\dots$ percent, which rounds to 28.3 percent.

28.3

Common wrong answer: dividing the total by the part instead of the part by the total gives $\frac{60}{17} \times 100 \approx 352.9$ percent — a percent above 100 for a part of the flock, which is the tell.

11. An 84 dollar deposit is 20% of a bicycle's price. Find the price and the amount still owed.

Solution: (a) $0.20x = 84$, so $x = 84 \div 0.20 = 420$ dollars.

420

(b) The deposit is already paid, so what remains is $420 - 84 = 336$ dollars.

336

12. 45 students are 30% of those asked. Solve for the number asked, then explain the division shortcut.

Solution: (a) $\frac{30}{100}x = 45$ becomes $0.30x = 45$, so $x = 45 \div 0.30 = 150$ students.

150

(b) **Instructor-judged.** The equation says the unknown whole was multiplied by 0.30 to produce 45. To undo a multiplication you divide by the same factor, so $45 \div 0.30$ *is* the solving step — written without bothering to name the unknown. Full credit needs that idea (dividing undoes the multiplying), in the student's own words, not merely the observation that both routes give 150.
